

When Two Proportions Are Not Independent

AP Statistics · Unit 3 · Topic 3.9 · B: 2026-12-10 · E: 2027-01-20 · TEACHER KEY

CED TETHER

- **UNC-3.N** Determine parameters of a sampling distribution for a difference in sample proportions. [Skill 3.B]
- **UNC-3.N.1** For a categorical variable, when randomly sampling from two independent populations with proportions p_1 and p_2 , the sampling distribution of the difference in sample proportions has mean $\mu = p_1 - p_2$.
- **UNC-3.N.2** If sampling without replacement and sample sizes are less than 10% of population sizes, the difference is negligible.
- **UNC-3.O** Determine whether a sampling distribution for a difference of sample proportions can be described as approximately normal. [Skill 3.C]
- **UNC-3.O.1** The sampling distribution of the difference in sample proportions will have an approximate normal distribution provided sample sizes are large enough: $n_1p_1 \geq 10$, $n_1(1 - p_1) \geq 10$, $n_2p_2 \geq 10$, $n_2(1 - p_2) \geq 10$.
- **UNC-3.P** Interpret probabilities and parameters for a sampling distribution for a difference in proportions. [Skill 4.B]
- **UNC-3.P.1** Parameters for a sampling distribution for a difference of proportions should be interpreted using appropriate units and within the context of specific populations.

Essential question: When is the independent two-proportion sampling model warranted, and what changes for paired observations?

DOK-3 skill: 4.B · **FRQ pattern:** independent-vs-paired-which-sampling-distribution-model-is-warranted

DOK rationale: Part (c) weighs model fit against design purpose, uses numerical condition evidence, distinguishes a useful paired plan from an independent analysis, and traces the consequence of omitting within-person association.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask students to mark one sound idea and one overreach across the two claims.
- Begin the lesson video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and attach every condition to the formula feature it supports.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose whose use of the formula is warranted and cite one design detail before the lesson.
- Complete the follow-along about center, spread, shape, and probabilities for differences in sample proportions.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What random object is selected in Plan I, and what is selected in Plan P?
- Which four expected counts establish the normal shape for Plan I?
- Does using 120 twice make the two results independent?
- What information about within-cardholder changes disappears if the plans are treated alike?

Adult role

Solo teacher. Circulate 5 pairs. Require students to separate a useful paired design from an invalid use of the independent-sample variance formula.

[WARN] WATCH FOR

- Treating equal sample sizes as evidence of independence.
- Subtracting the two variance terms because the proportions are subtracted.
- Calling Plan P unusable instead of identifying the need for a paired analysis.

SCORING PART (c) — E / P / I

Expected elements

- **validates-plan-i-model** (required) — Applies the independent model to Plan I and cites separate selection plus both 10 percent checks and all four passing counts
- **recognizes-paired-design-value** (required) — Recognizes that following the same people can control stable person-to-person differences or target within-person change
- **diagnoses-pairing** (required) — Explains that Plan P links each person's outcomes and that equal sample sizes do not imply independence
- **states-model-consequence** (required) — Explains that omitting within-person association can distort the 0.0607 spread and 0.0350 probability
- **preserves-paired-information** (required) — Describes a valid Plan P analysis that retains individual pairs or discordant transitions

E	Correctly validates Plan I with numerical conditions, recognizes Plan P's potential design value, diagnoses its dependence, explains the uncertainty consequence, and preserves paired information in a valid alternative analysis.
P	Correctly separates the independent and paired structures but omits one numerical condition set, Mateo's defensible design point, a specific consequence, or what the paired analysis must retain.
I	Transfers the independent formula to Plan P because sample sizes match, ignores the repeated people, or dismisses Plan P rather than changing the analysis.

Common mistakes

- Treating equal sample sizes as an independence condition
- Subtracting the variance contributions because the proportions are subtracted
- Using the independent-sample normal probability for the same 120 people
- Calling the paired design invalid rather than changing the analysis

[STAR] TODAY'S PROBLEM — annotated

A library studies self-checkout before and after a redesign. Records give $p_B = 0.42$ among 4,000 before visits and $p_A = 0.54$ among 5,000 after visits.

Plan I: Draw an SRS of 120 before visits and a separate SRS of 150 after visits; do not track cardholders.

Plan P: Measure the same 120 cardholders before and after.

Sora: Separate selection warrants Plan I's independent model; 0.0350 means a difference at most 0.01 is unusual.

Mateo: Pairing controls person differences, so Plan P may help. Because both proportions use 120 people, Plan I's model applies to P too.

Sampling plans

Plan	Before	After	Repeated
I	SRS 120	SRS 150	no
P	120	same 120	yes

FIRST TAKE (5 min)

Which parts of Sora's and Mateo's reasoning are warranted? Cite one design detail.

Key: Not graded. Sora matches the model; Mateo values a sound design but transfers the wrong model.

Rules callout "THE VARIANCE FORMULA REQUIRES INDEPENDENT SAMPLES (keep on the page)" is printed on the student sheet.

DOK 1 (a) For after minus before, identify the parameter and statistic and state Plan I's sampling-distribution mean.

Key: The parameter is $p_A - p_B$, the difference between the after- and before-period population proportions of visits using self-checkout. The statistic is $\hat{p}_A - \hat{p}_B$. For Plan I, $\mu_{\hat{p}_A - \hat{p}_B} = p_A - p_B = 0.54 - 0.42 = 0.12$.

DOK 2 (b) For Plan I, check both 10 percent conditions and all four counts. Find the standard deviation and approximate $P(\hat{p}_A - \hat{p}_B \leq 0.01)$; interpret it.

Key: Both 10 percent checks hold: $150 < 500$ and $120 < 400$; the four counts are 81, 69, 50.4, and 69.6. The variance is $0.001656 + 0.002030 = 0.003686$, so $SD = \sqrt{0.003686} = 0.0607124$. Then $z = -1.8118$ and $P(\hat{p}_A - \hat{p}_B \leq 0.01) \approx 0.0350$. About 3.5 percent of independent sample pairs would show a difference at most 0.01 when the population difference is 0.12.

DOK 3 (c) Adjudicate the claims. Use design and conditions to place the independent model and assess Plan P. Explain why equal sample sizes are not independence, how transferring 0.0607 and 0.0350 could mislead, and what paired analysis must retain.

Key: Sora is correct for Plan I: separate SRSs and passing conditions warrant 0.0607 and 0.0350. Mateo is right that pairing can control stable person differences. But Plan P links each person's outcomes; equal sample sizes do not imply independence. The independent formula omits within-person association and could distort spread and tail probability in either direction. A valid paired analysis keeps each pair or the discordant transitions. Plan I fits the independent model; Plan P can be useful with paired analysis.

EXIT

One thing the video changed about my first take: _____